

Report
Of
**"Java Full Stack Development With Project Virtual
Internship"**

Submitted To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By
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Batch: 2026-27

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CANDIDATE'S DECLARATION

I, Shashwat Upadhyay, Roll Number 2410030445, do hereby declare that the internship report titled " **Java Full Stack Development With Project Virtual Internship** " has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Student Name: Shashwat Upadhyay

Roll No.: 2410030445

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to extend my sincere thanks to EduSkills Academy and AICTE for providing me the opportunity to undertake this 8-week Virtual Internship Program in Java Full Stack Development with Project, under the National Internship Portal, Ministry of Education.

I express my gratitude to my parents for their blessings.

Student Name: Shashwat Upadhyay

Roll No.: 2410030445

INTERNSHIP COMPLETION CERTIFICATE



शिक्षा मंत्रालय
MINISTRY OF
EDUCATION



AICTE
अखिल भारतीय तकनीकी शिक्षा परिषद्
All India Council for Technical Education



NATIONAL
INTERNSHIP
PORTAL



EduSkills®
Nation Building Through Skills



Certificate of Virtual Internship

This is to certify that

Shashwat Upadhyay

IILM University, Greater Noida

has successfully completed the 8-weeks

Java Full Stack Development With Project Virtual Internship

During June - August 2026

Supported by  **EduSkills**
ACADEMY



Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education



Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills



Certificate ID : 4369910339a7fa279ce0
Student ID: STU699786ca2a9381771538122



GRADE: O (Outstanding): 90-100 | E (Excellent): 80-89 | A (Very Good): 70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

4. PROJECT DESCRIPTION

4.1 Introduction

Industry-oriented learning through internships plays a crucial role in bridging the gap between academic knowledge and real-world application. As part of the AICTE – EduSkills Virtual Internship Program, an 8-week internship was undertaken in the domain of Java Full Stack Development, during June 2026 to August 2026. The internship provided hands-on exposure to front-end and back-end web technologies (HTML, CSS, Bootstrap, JavaScript/jQuery, Core and Advanced Java, Spring Framework, Spring Boot, Hibernate, MySQL and Git) and their application in modern software solutions, culminating in a project titled “Text Search Indexer.”

Internship Overview:

- Internship Mode: Online / Virtual
- Duration: 8 Weeks (June 2026 – August 2026)
- Platform Used: EduSkills LMS
- Role: Intern
- Domain: Java Full Stack Development + Project

4.2 Organization Profile

Organization Name: EduSkills Academy

Domain: EdTech & Skill Development

Services: Virtual Internships, delivered under the AICTE National Internship Portal in association with the Ministry of Education, Government of India.

Vision: To empower students with industry-relevant skills through structured, mentor-guided virtual internship programs.

4.3 Problem Statement

The Text Search Indexer is a Java-based application designed to create an index for text documents, enabling efficient searching. The project encompasses several key stages of search-engine development, beginning with the preprocessing of textual data. It includes robust text normalization to standardize input, followed by tokenization to break down text into manageable units. The system further identifies unique terms and

calculates term frequencies to build a foundational index. Core functionality is augmented by a keyword-matching component that allows basic search queries. The project aims to provide practical insight into the internal workings of text search systems, from data ingestion to retrieval.

4.4 Project Objectives

- To design and implement a working solution for the given problem statement.
- To utilise modern Java tools, data structures and algorithms for text processing.
- To optimise the performance and quality of the code developed for each module.

4.5 Scope of the Project

The present work is scoped to processing a single text document at a time and implementing the foundational building blocks of a search engine — normalization, tokenization, unique-word indexing, keyword matching, and term-frequency counting. The project can be extended further to support phrase searching and boolean operators (AND, OR, NOT), incorporate more sophisticated ranking algorithms such as TF-IDF to improve result relevance, and extend the indexer to search across multiple documents concurrently.

4.6 Technologies and Tools Used

- Programming Language: Java
- Core Java Libraries: Standard Java I/O, Collections Framework (HashMap, ArrayList, TreeSet), String manipulation methods
- Concepts Applied: Text processing, tokenization, normalization, data structures (maps and sets), frequency-counting algorithms, search-indexing fundamentals
- Platform: EduSkills LMS, used for weekly assessments and project submission

4.7 System Architecture

The Text Search Indexer follows a simple, linear pipeline architecture in which raw text passes sequentially through five independent processing modules, each consuming the previous module's output:

**Raw Text Input → Text Normalization → Tokenizer → Unique Word
Indexer → Keyword Search Matcher → Term Frequency Counter → JSON
Output**

Each module is implemented as a standalone Java console program that reads input line-by-line using Scanner and emits a structured JSON response, which keeps the modules loosely coupled and independently testable.

4.8 Methodology

The development of the Text Search Indexer was approached in a step-by-step, modular fashion, addressing each component of the search pipeline sequentially:

- Text Normalization Module – converts all input text to lowercase and removes non-alphanumeric characters, ensuring that variations in casing and punctuation do not affect downstream processing.
- Basic Tokenizer – splits normalized text on whitespace to produce a structured list of tokens for further analysis.
- Unique Word Indexer – processes the tokens using a TreeSet to identify and store distinct words, forming the vocabulary for the index.
- Keyword Search Matcher – checks a line of text for the presence of a specified keyword and returns a boolean result.
- Term Frequency Counter – uses a HashMap to count the occurrence of each word in a line, providing the basis for ranking search results.

As an illustration, the Text Normalization module was implemented as follows:

```
line.toLowerCase().replaceAll("[^a-z0-9 ]",  
"").replaceAll("\\s+", " ").trim();
```

Similar single-purpose console programs were written for the remaining four modules, each validated independently against sample inputs before being considered complete. The complete source code for all five modules is included with the project submission.

4.9 Expected Outcomes

- Practical experience in implementing core text-processing techniques in Java, including normalization and tokenization.
- A foundational understanding of data structures such as HashMap and TreeSet for building frequency counts and managing indexes.
- The ability to systematically approach complex problems by breaking them into smaller, manageable modules.
- Skills in designing and implementing algorithms for text analysis and search.

Overall, the Text Search Indexer project demonstrates the fundamental principles of building a text-processing and search system. By implementing modules for normalization, tokenization, indexing and basic search, the project lays a solid foundation for understanding how search engines process and retrieve information, and offers a practical example of applying Java programming to real-world data-analysis problems.

4.10 Certificates of Completion and Other Communication Proof

The following pages present the AICTE–EduSkills internship Offer Letter and the Certificate of Virtual Internship issued on successful completion of the program, as proof of communication and completion.

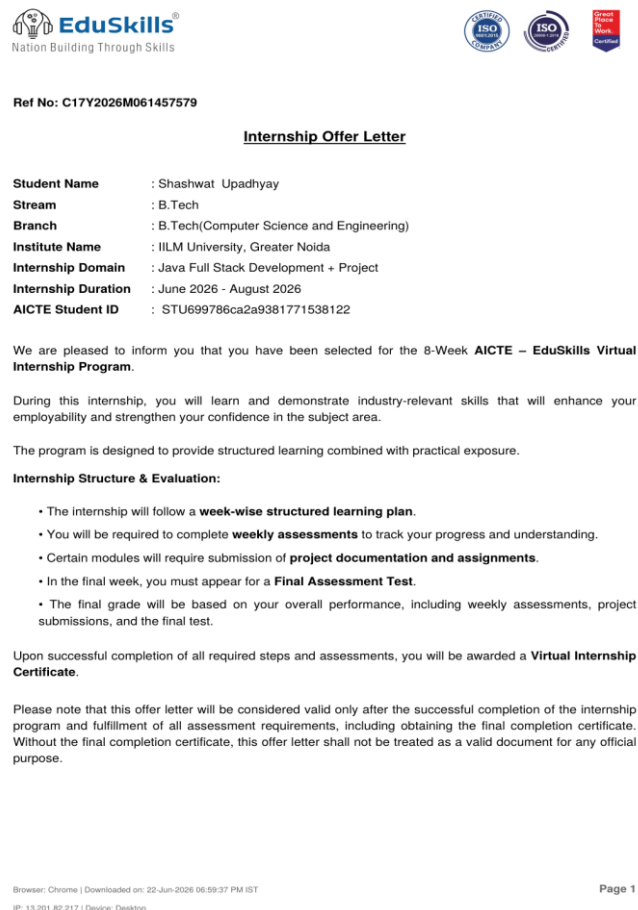


Fig. 4.10.1: Internship Offer Letter (Page 1)

The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	HTML	Learn HTML fundamentals including page structure, semantic elements, forms, tables, and building the foundation of web pages.
Week 2	CSS	Master CSS styling concepts including selectors, box model, positioning, flexbox, grid, and responsive web design.
Week 3	Bootstrap	Build responsive and mobile-first interfaces using Bootstrap components, grid system, and utility classes.
Week 4	JavaScript & jQuery	Learn JavaScript fundamentals, DOM manipulation, events, and enhance development efficiency using jQuery for interactive web applications.
Week 5	Core Java	Understand Java fundamentals including OOP concepts, data types, collections, exception handling, and file handling.
Week 6	Advanced Java	Explore advanced Java concepts including JDBC, Servlets, multithreading, and enterprise-level application development principles.
Week 7	Spring Framework & Spring Boot	Develop enterprise applications using Spring Framework and Spring Boot, covering dependency injection, REST APIs, and application configuration.
Week 8	Hibernate, MySQL & Git	Learn ORM concepts using Hibernate and integrate MySQL database for data persistence and CRUD operations. Manage source code using Git and GitHub, implement branching strategies, and collaborate effectively in team environments.
Project / Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.



Fig. 4.10.2: Internship Offer Letter (Page 2 – Weekly Module Structure)



Fig. 4.10.3: Certificate of Virtual Internship

5. BIBLIOGRAPHY/REFERENCES

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